6.1200J/18.062J Mathematics for Computer Science
Problem Set Solutions

Nouman Rahman

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Contents

Part I
Problem Set 1

Problem 1: Collaboration Policy

Problem 2: Set Formulas and Propositional Formulas

(a) Verify that $(P \wedge \bar{Q}) \vee (P \wedge Q)$ is equivalent to P using a truth table.

P	Q	$\bar{Q}$	$P \wedge \bar{Q}$	$P \wedge Q$	$(P \wedge \bar{Q}) \vee (P \wedge Q)$	P
T	T	F	F	T	T	T
T	F	T	T	F	T	T
F	T	F	F	F	F	F
F	F	T	F	F	F	F

The column for $(P \wedge \bar{Q}) \vee (P \wedge Q)$ agrees with the column for P in every row, so the two formulas are equivalent.

(b) Prove that $A = (A - B) \cup (A \cap B)$ for all sets A, B, by showing $x \in A$ iff $x \in (A - B) \cup (A \cap B)$ for all elements x.

Proof. Let x be arbitrary.

($\Rightarrow$) Suppose $x \in A$. Either $x \in B$ or $x \notin B$.

- If $x \notin B$, then $x \in A$ and $x \notin B$, so $x \in A - B$, hence $x \in (A - B) \cup (A \cap B)$.
- If $x \in B$, then $x \in A$ and $x \in B$, so $x \in A \cap B$, hence $x \in (A - B) \cup (A \cap B)$.

Either way, $x \in (A - B) \cup (A \cap B)$.

($\Leftarrow$) Suppose $x \in (A - B) \cup (A \cap B)$. Then $x \in A - B$ or $x \in A \cap B$.

- If $x \in A - B$, then by definition $x \in A$ (and $x \notin B$).
- If $x \in A \cap B$, then $x \in A$ (and $x \in B$).

Either way, $x \in A$.

Since $x \in A$ iff $x \in (A - B) \cup (A \cap B)$ for arbitrary x, the two sets are equal: $A = (A - B) \cup (A \cap B)$. $\square$

Relation to part (a). This is the set-theoretic mirror of part (a). Substitute $P :=$ "$x \in A$" and $Q :=$ "$x \in B$". Then:

- $x \in A - B$ corresponds to $P \wedge \bar{Q}$ (in A, not in B),
- $x \in A \cap B$ corresponds to $P \wedge Q$ (in A, in B),
- $\cup$ corresponds to $\vee$.

So the set identity $A = (A - B) \cup (A \cap B)$ is exactly the membership-level statement $P \leftrightarrow \big((P \wedge \bar{Q}) \vee (P \wedge Q)\big)$ from part (a), and the case analysis in the proof above is just an unfolding of the truth table: the two cases $x \notin B$ / $x \in B$ are the two ways Q can be false or true, matching the two rows of the truth table where P is true.

Problem 3: Predicate Practice

(a) The predicate sixish(n), "n is a natural number whose first digit is 6":

$$\text{sixish}(n) := \ \exists k \in \mathbb{N}.\ \big(6 \cdot 10^k \leq n < 7 \cdot 10^k\big).$$

(b) "There are exactly two natural numbers n such that $Q(n)$ is true":

$$\exists m, n \in \mathbb{N}.\ \Big(m \neq n\ \wedge\ \forall k \in \mathbb{N}.\big(Q(k) \leftrightarrow (k = m \vee k = n)\big)\Big).$$

(c) Goldbach's Conjecture: "every even integer greater than 2 can be written as the sum of two primes." Following the required form $\forall n.\ W(n)$ with all the logic folded into W:

$$\forall k \in \mathbb{N}.\ \Big[\big(k > 2 \wedge \text{isEven}(k)\big) \rightarrow \exists m, n \in \mathbb{N}.\big(\text{isPrime}(m) \wedge \text{isPrime}(n) \wedge k = m + n\big)\Big],$$

where, since isEven was not predefined for us, we take $\text{isEven}(k) := \neg\text{isOdd}(k)$.

(d) "Every prime larger than 100 is sixish":

$$\forall n \in \mathbb{N}.\ \Big[\big(n > 100 \wedge \text{isPrime}(n)\big) \rightarrow \text{sixish}(n)\Big].$$

(e) Is $R(n) := \exists b \in \mathbb{N}.\ (n = 2b + 5)$ equivalent to $\text{isOdd}(n)$?
Claim: No, R is <u>not</u> equivalent to isOdd.

Proof. Suppose toward a contradiction that $R(n) \leftrightarrow \text{isOdd}(n)$ for all $n \in \mathbb{N}$. Take $n = 3$. Since 3 is odd, isOdd(3) is true, so by our assumption $R(3)$ must also be true. But

$$R(3) := \exists b \in \mathbb{N}.\ (3 = 2b + 5),$$

and solving $3 = 2b + 5$ gives $b = -1 \notin \mathbb{N}$, so no such b exists and $R(3)$ is false. This contradicts $R(3)$ being true. Hence our assumption was wrong, and R is not equivalent to isOdd.

(Intuitively, $R(n)$ only captures odd numbers $n \geq 5$, since b ranges over $\mathbb{N}$; it misses the smaller odd numbers 1 and 3.) $\square$

Problem 4: Induction and Contradiction

(a) *Setup.* $Q(x)$ is a predicate on $\mathbb{R}$ such that for all $x, y \in \mathbb{R}$, if $Q(x \cdot y)$ is true then $Q(x)$ or $Q(y)$ is true. We show the stronger property: for every integer $n \geq 1$ and all $x_1, \ldots, x_n \in \mathbb{R}$, if $Q(x_1 x_2 \cdots x_n)$ is true then at least one of $Q(x_1), \ldots, Q(x_n)$ is true.

Proof. Let $P(n)$ be the statement: for all $x_1, \ldots, x_n \in \mathbb{R}$,

$$Q(x_1 x_2 \cdots x_n) \rightarrow Q(x_1) \vee Q(x_2) \vee \cdots \vee Q(x_n).$$

We prove $P(n)$ holds for all integers $n \geq 1$ by induction on n.

Base case ($n = 1$): $P(1)$ says: for all $x_1 \in \mathbb{R}$, $Q(x_1) \rightarrow Q(x_1)$, which is trivially true.

Inductive step: Suppose $n \geq 1$ and $P(n)$ holds; we show $P(n+1)$ holds. Let $x_1, \ldots, x_n, x_{n+1} \in \mathbb{R}$ be arbitrary, and suppose $Q(x_1 x_2 \cdots x_n x_{n+1})$ is true. Write

$$y := x_1 x_2 \cdots x_n, \qquad z := x_{n+1},$$

so that $x_1 \cdots x_n x_{n+1} = yz$. By hypothesis, $Q(yz)$ is true. Applying the given property of Q (i.e., $P(1)$ is not what we need here — we use the defining property of Q directly, which is exactly $P(2)$ applied to y, z) to y and z:

$$Q(yz) \to Q(y) \vee Q(z).$$

Since $Q(yz)$ holds, we conclude $Q(y) \vee Q(z)$ holds, i.e.,

$$Q(x_1 x_2 \cdots x_n) \ \vee \ Q(x_{n+1}).$$

Now consider the two cases:

- If $Q(x_{n+1})$ holds, then in particular $Q(x_1) \vee \cdots \vee Q(x_n) \vee Q(x_{n+1})$ holds.
- If $Q(x_1 x_2 \cdots x_n)$ holds, then by the inductive hypothesis $P(n)$ applied to $x_1, \ldots, x_n$, we get

$$Q(x_1 x_2 \cdots x_n) \to Q(x_1) \vee \cdots \vee Q(x_n),$$

  so $Q(x_1) \vee \cdots \vee Q(x_n)$ holds, and hence so does $Q(x_1) \vee \cdots \vee Q(x_n) \vee Q(x_{n+1})$.

In either case, $Q(x_1) \vee \cdots \vee Q(x_n) \vee Q(x_{n+1})$ holds, which is exactly $P(n+1)$.

By mathematical induction, $P(n)$ holds for all integers $n \geq 1$. □

Remark (fixing a gap). The original argument substituted y and z back into $Q(y) \vee Q(z)$ and asserted the result followed "by the inductive hypothesis," but never explicitly invoked $P(n)$ on $Q(y) = Q(x_1 \cdots x_n)$. The step above makes that invocation explicit: we need $P(n)$ to expand $Q(x_1 \cdots x_n)$ into $Q(x_1) \vee \cdots \vee Q(x_n)$.

(b) *Theorem 1.* If the product of $n \geq 1$ real numbers is irrational, then at least one of those numbers must be irrational.

To derive this from part (a), define

$$Q(x) := \text{“}x \text{ is irrational.”}$$

Before part (a) applies to this Q, we must check Q satisfies the hypothesis of part (a), namely the base property

$$\forall x, y \in \mathbb{R}. \ \big(Q(xy) \to Q(x) \vee Q(y)\big).$$

Equivalently (by contrapositive), we must prove the following lemma:

> **Lemma.** For all $x, y \in \mathbb{R}$, if x and y are both rational, then xy is rational.

Once the Lemma is established, part (a) (with this Q) gives exactly Theorem 1.

(c) *Proof of the Lemma, by contradiction.*

Proof. Suppose, toward a contradiction, that x and y are both rational but xy is irrational. Since x and y are rational, by definition there exist $p_1, q_1, p_2, q_2 \in \mathbb{Z}$ with $q_1 \neq 0$, $q_2 \neq 0$, such that

$$x = \frac{p_1}{q_1}, \qquad y = \frac{p_2}{q_2}.$$

Then

$$xy = \frac{p_1}{q_1} \cdot \frac{p_2}{q_2} = \frac{p_1 p_2}{q_1 q_2}.$$

Since $p_1, p_2, q_1, q_2 \in \mathbb{Z}$, we have $p_1 p_2 \in \mathbb{Z}$ and $q_1 q_2 \in \mathbb{Z}$; moreover $q_1 q_2 \neq 0$ since $q_1 \neq 0$ and $q_2 \neq 0$. Hence xy is expressible as a ratio of two integers with nonzero denominator, i.e. xy is rational. This contradicts our assumption that xy is irrational. Therefore, no such x, y exist: whenever x and y are both rational, xy is rational. □

Finishing the proof of Theorem 1. The Lemma is precisely the contrapositive of the required property of $Q(x) :=$ "x is irrational," namely

$$\forall x, y \in \mathbb{R}.\ Q(xy) \to Q(x) \vee Q(y).$$

(Indeed: if xy is irrational, then x, y cannot both be rational, by the Lemma; so at least one of x, y is irrational.) Since Q satisfies the hypothesis of part (a), part (a) tells us that for every integer $n \geq 1$ and all $x_1, \ldots, x_n \in \mathbb{R}$,

$$Q(x_1 x_2 \cdots x_n) \to Q(x_1) \vee \cdots \vee Q(x_n),$$

i.e., if $x_1 x_2 \cdots x_n$ is irrational, then at least one of $x_1, \ldots, x_n$ is irrational. This is exactly Theorem 1. ■

Part II
Problem Set 2

Problem 1: Outlining Proofs

(a) Theorem: "Every natural number is tall or wide, but not both." (Proof method: proof by cases.)

Writing $T(n)$ for "n is tall" and $W(n)$ for "n is wide," the theorem says

$$\forall n \in \mathbb{N}.\ \big(T(n) \oplus W(n)\big),$$

where $\oplus$ is exclusive or. We outline a proof by cases on whether n is tall or wide (this case split itself needs $T(n) \vee W(n)$ to be already known, e.g. from an earlier result — we flag this as a TODO as well).

Proof Outline. Let $n \in \mathbb{N}$ be arbitrary. *[TODO: show $T(n) \vee W(n)$ (needed so the two cases below are exhaustive)].* We consider two cases.

Case 1: $T(n)$ holds. We must show $T(n) \oplus W(n)$, i.e. (since $T(n)$ is true) that $W(n)$ is false. *[TODO: show $T(n) \to \neg W(n)$].*

Case 2: $W(n)$ holds. Symmetrically, we must show $T(n)$ is false. *[TODO: show $W(n) \to \neg T(n)$].*

In both cases $T(n) \oplus W(n)$ holds, completing the proof by cases.

(b) Theorem: "There exists a happy natural number between 10 and 1000 that has no happy proper divisors." (Proof method: choose 25 as the example.)

Writing $H(n)$ for "n is happy" and $D(n)$ for the set of proper divisors of n, the theorem says

$$\exists n \in \mathbb{N}.\ \Big(10 \le n \le 1000\ \wedge\ H(n)\ \wedge\ \forall d.\big(d \mid n \wedge d < n \to \neg H(d)\big)\Big).$$

Proof Outline. Take $n = 25$.

$10 \le 25 \le 1000$: true by inspection.

[TODO: show $H(25)$, i.e. that 25 is happy].

The proper divisors of 25 are 1 and 5. *[TODO: show $\neg H(1)$ and $\neg H(5)$].*

Since 25 is in range, happy, and has no happy proper divisor, $n = 25$ witnesses the existential claim.

(c) Theorem: "There are no devious natural numbers." (Proof method: regular induction, with base cases 0 and 1.)

Writing $D(n)$ for "n is devious," the theorem says $\forall n \in \mathbb{N}.\ \neg D(n)$. Let

$$P(n) := \neg D(n).$$

Two base cases are called for, which suggests the (unstated) recursive structure of "devious" relates n to its two predecessors $n-1, n-2$ (as in Problem 2). We outline accordingly:

Proof Outline. We prove $P(n)$ for all $n \in \mathbb{N}$ by induction on n, with two base cases and a two-term inductive step.

Base case $n = 0$**:** *[TODO: show $P(0)$, i.e. $\neg D(0)$].*

Base case $n = 1$**:** *[TODO: show $P(1)$, i.e. $\neg D(1)$].*

Inductive step: Let $k \geq 1$, and assume $P(k-1)$ and $P(k)$ (i.e. $\neg D(k-1)$ and $\neg D(k)$) both hold. *[TODO: show $P(k+1)$, i.e. $\neg D(k+1)$, using the definition of devious and the two inductive hypotheses].*

By induction (using both base cases as anchors), $P(n)$ holds for all $n \in \mathbb{N}$.

Problem 2: Spotting Errors

Recall $x_1 = 4$, $x_2 = 13$, and $x_i = 6x_{i-1} - 8x_{i-2}$ for $i \geq 3$, so $(x_1, x_2, x_3, x_4, \dots) = (4, 13, 46, 172, \dots)$. Goal: $x_i > 0$ for all $i \geq 1$.

(a) Befuddled Brynmor's mistake.

Brynmor uses strong induction on $Q(i) :=$ "$x_i \geq 2^i$", and in the inductive step writes

$$x_{i+1} = 6x_i - 8x_{i-1} \ \geq \ 6 \cdot 2^i - 8 \cdot 2^{i-1},$$

substituting the lower bounds $x_i \geq 2^i$ and $x_{i-1} \geq 2^{i-1}$ directly into the expression.

This step is invalid. The term $6x_i$ has a positive coefficient, so substituting the lower bound $x_i \geq 2^i$ correctly gives $6x_i \geq 6 \cdot 2^i$. But the term $-8x_{i-1}$ has a *negative* coefficient: to lower-bound $-8x_{i-1}$ we need an *upper* bound on x_{i-1}, not a lower bound. Knowing $x_{i-1} \geq 2^{i-1}$ only tells us $-8x_{i-1} \leq -8 \cdot 2^{i-1}$ — the inequality goes the wrong way to conclude anything about x_{i+1} being large. So the displayed inequality $x_{i+1} \geq 6 \cdot 2^i - 8 \cdot 2^{i-1}$ does not actually follow from the inductive hypothesis, and the rest of the argument is unjustified.

(b) Erroneous Erik's mistake.

Erik uses (regular) induction on $R(i) :=$ "$x_i \geq 3x_{i-1}$" for $i \geq 2$. In the inductive step, after correctly deriving $x_{i+1} \geq \frac{10}{3}x_i$ from the inductive hypothesis $R(i)$, he writes: "Since $x_i > 0$, $\frac{10}{3}x_i > 3x_i$" in order to conclude $x_{i+1} \geq 3x_i$, i.e. $R(i+1)$.

This step is circular. The inductive step for $R(i+1)$ is only allowed to use the inductive hypothesis $R(i)$ (namely $x_i \geq 3x_{i-1}$) — it does *not* give us $x_i > 0$ for free. Positivity of x_i is exactly the fact the whole exercise is trying to establish, and it is only derived *afterward*, by combining $R(i)$ with $x_1 > 0$. Using "$x_i > 0$" inside the proof of $R(i) \Rightarrow R(i+1)$, before positivity has been established, assumes the very conclusion the argument is building toward.

(c) Overzealous Zach's mistake.

Zach uses strong induction on $S(i) :=$ "$x_i = 4^i$", with a single base case $S(1)$, and an inductive step that (for general $i \geq 1$) invokes both $S(i-1)$ and $S(i)$ to derive $S(i+1)$ via the recurrence $x_{i+1} = 6x_i - 8x_{i-1}$.

This step fails already at $i = 1$. To derive $S(2)$ this way, the inductive step needs $S(0)$, i.e. a value $x_0 = 4^0 = 1$ — but x_0 is not defined at all (x_1, x_2 are the given base values, and the recurrence $x_i = 6x_{i-1} - 8x_{i-2}$ is only stipulated for $i \geq 3$). So the inductive step's use of the recurrence to compute x_2 from x_1, x_0 is illegitimate on two counts: x_0 doesn't exist, and even x_2 itself is a *given* base value, not something produced by the recurrence. Consistent with this, the claim $S(2)$ is simply false: $x_2 = 13 \neq 16 = 4^2$. So the strong induction breaks down exactly at the step from $i = 1$ to $i = 2$.

(d) Correct proof that $x_i > 0$ for all $i \geq 1$.

Following the hint, we repair Erik's argument by folding positivity directly into the inductive hypothesis, so it no longer needs to be smuggled in mid-proof. Let

$$P(i) := \text{"}x_i > 2x_{i-1}\text{"}, \qquad \text{for } i \geq 2.$$

Proof. We prove $P(i)$ holds for all $i \geq 2$ by induction on i.

Base case ($i = 2$): $x_2 = 13$ and $2x_1 = 8$, and indeed $13 > 8$, so $P(2)$ holds.

Inductive step: Let $i \geq 2$ and assume $P(i)$, i.e. $x_i > 2x_{i-1}$. We show $P(i+1)$, i.e. $x_{i+1} > 2x_i$.

From $x_i > 2x_{i-1}$ we get $x_{i-1} < \frac{1}{2}x_i$. Multiplying both sides by -8 (which flips the inequality, since $-8 < 0$):

$$-8x_{i-1} > -8 \cdot \tfrac{1}{2}x_i = -4x_i.$$

Now use the recurrence $x_{i+1} = 6x_i - 8x_{i-1}$:

$$x_{i+1} = 6x_i - 8x_{i-1} \ > \ 6x_i - 4x_i = 2x_i.$$

So $x_{i+1} > 2x_i$, which is $P(i+1)$.

By induction, $P(i)$ holds for all $i \geq 2$, i.e.

$$x_i > 2x_{i-1} \qquad \text{for all } i \geq 2. \tag{$*$}$$

Finally, we conclude $x_i > 0$ for all $i \geq 1$ by a second, simple induction, using $(*)$:

- $x_1 = 4 > 0$.
- If $x_{i-1} > 0$ for some $i \geq 2$, then by $(*)$, $x_i > 2x_{i-1} > 2 \cdot 0 = 0$.

Hence $x_i > 0$ for every $i \geq 1$, as required. □

Problem 3: Integer Multiplication from Bit Shifts

State machine on triples $(r, s, a) \in \mathbb{N}^3$, start state $(x, y, 0)$, transitions (defined only when $s > 0$):

$$(r, s, a) \mapsto \begin{cases} (2r,\ s/2,\ a) & \text{if } s \text{ is even (and } s > 0) \\ (2r,\ (s-1)/2,\ a+r) & \text{if } s \text{ is odd} \end{cases}$$

(a) Tracing from $(5, 22, 0)$ (target: $5 \cdot 22 = 110$):

$$(5, 22, 0) \to (10, 11, 0) \to (20, 5, 10) \to (40, 2, 30) \to (80, 1, 30) \to (160, 0, 110).$$

No transition applies once $s = 0$, so this is the final state, with $a_f = 110 = 5 \cdot 22$. It correctly computes the product.

(b) Invariant. Let $P(r, s, a) :=$ "$rs + a = xy$." We show P is an invariant of the state machine: true at the start state, and preserved by every transition.

Proof. **Holds initially:** $P(x, y, 0)$ says $xy + 0 = xy$, which is true.

Preserved by transitions: Suppose $P(r, s, a)$ holds, i.e. $rs + a = xy$, and $s > 0$ so a transition applies.

Case s even: $(r, s, a) \mapsto (2r,\ s/2,\ a)$. Then

$$(2r)\left(\frac{s}{2}\right) + a = rs + a = xy,$$

so $P(2r, s/2, a)$ holds.

Case s odd: $(r, s, a) \mapsto (2r,\ (s-1)/2,\ a+r)$. Then

$$(2r)\left(\frac{s-1}{2}\right) + (a + r) = r(s-1) + a + r = rs - r + a + r = rs + a = xy,$$

so $P(2r, (s-1)/2, a+r)$ holds.

In both cases, the successor state also satisfies P. Since P holds at the start state and is preserved by every transition, P is an invariant. □

(c) Final states. Transitions are defined exactly when $s > 0$, so a state has no outgoing transition (is final) precisely when $s = 0$. Thus the final states are exactly the triples of the form $(r, 0, a)$ for $r, a \in \mathbb{N}$.

(d) Partial correctness. Let (r_f, s_f, a_f) be any reachable final state. By part (c), $s_f = 0$. Since P is an invariant (part (b)) and holds at the start state, it holds at every reachable state, including this one:

$$P(r_f, 0, a_f): \qquad r_f \cdot 0 + a_f = xy \quad \Longrightarrow \quad a_f = xy.$$

So in any reachable final state, $a_f = xy$, as required.

(e) Termination. Let $\mathrm{bitcount}(n) := \lceil \log_2(n+1) \rceil$ for $n \in \mathbb{N}$. We show $\mathrm{bitcount}(s)$ strictly decreases — in fact, decreases by exactly 1 — with every transition.

It is a standard fact that for $n \geq 1$, $\mathrm{bitcount}(n)$ is the unique integer k such that $2^{k-1} \leq n < 2^k$ (i.e. the number of bits in n's binary representation), and $\mathrm{bitcount}(0) = 0$.

Proof. Let (r, s, a) be any state with $s > 0$, and suppose $\text{bitcount}(s) = k$, so $2^{k-1} \leq s < 2^k$.

Case s odd: the transition sends $s \mapsto \dfrac{s-1}{2}$. Write $m = \dfrac{s-1}{2}$, so $s = 2m+1$. From $2^{k-1} \leq s < 2^k$ we get $2^{k-1} \leq 2m+1 < 2^k$, i.e. $2^{k-1} - 1 \leq 2m < 2^k - 1$; since $2m$ is even and $2^{k-1} - 1$ is odd (for $k \geq 1$), this sharpens to $2^{k-1} \leq 2m \leq 2^k - 2$, i.e. $2^{k-2} \leq m \leq 2^{k-1} - 1 < 2^{k-1}$. If $k \geq 2$ this says $2^{(k-1)-1} \leq m < 2^{k-1}$, so $\text{bitcount}(m) = k - 1$. (If $k = 1$, then $s = 1$, $m = 0$, and $\text{bitcount}(0) = 0 = k - 1$ by definition.) Either way,

$$\text{bitcount}\left(\frac{s-1}{2}\right) = k - 1 = \text{bitcount}(s) - 1.$$

Case s even: the transition sends $s \mapsto \dfrac{s}{2}$. Write $m = \dfrac{s}{2}$, so $s = 2m$, and (since $s > 0$) $m \geq 1$. From $2^{k-1} \leq s < 2^k$ we get $2^{k-1} \leq 2m < 2^k$, i.e. $2^{k-2} \leq m < 2^{k-1}$, so $\text{bitcount}(m) = k - 1$. Hence

$$\text{bitcount}\left(\frac{s}{2}\right) = k - 1 = \text{bitcount}(s) - 1.$$

In both cases, the new value of s has bitcount exactly one less than before. In particular, $\text{bitcount}(s)$ is a strictly decreasing, non-negative-integer-valued derived variable along any run of the machine. Since it cannot decrease forever (it is a non-negative integer), the machine cannot run forever: it must reach a state with $s = 0$ (no further transition possible), i.e. the algorithm always terminates. $\square$

(f) Time bound. By part (e), $\text{bitcount}(s)$ decreases by exactly 1 at every step, starting from $\text{bitcount}(y)$ at the initial state $(x, y, 0)$ and ending at $\text{bitcount}(0) = 0$ at the final state. Hence, the number of transitions taken is exactly

$$\text{bitcount}(y) = \lceil \log_2(y+1) \rceil,$$

so from any starting state $(x, y, 0)$, the algorithm terminates after at most $\lceil \log_2(y+1) \rceil$ steps.

Problem 4: Invariant Enlightenment